

# **Drug Utilisation Studies in the OMOP Common Data Model**

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# Preface

## Is this book for me?

This book is part of the collection of **Tidy R programming in OMOP** see accompanying books:

- (1) Tidy R programming with the OMOP common data model
- (2) Tidy R packages with the OMOP common data model
- (3) Drug Utilisation Studies in the OMOP Common Data Model

We've written this book for anyone interested to conduct DrugUtilisation studies using the OMOP Common Data Model and working with databases using a tidyverse style approach. That is, human centered, consistent, composable, and inclusive (see <https://design.tidyverse.org/unifying.html> for more details on these principles).

New to R? We recommend you compliment the book with R for data science

New to databases? We recommend you take a look at some web tutorials on SQL, such as SQLBolt or SQLZoo

New to the OMOP CDM? We'd recommend you pare this book with The Book of OHDSI

New to the Tidy R in OMOP ecosystem? Probably you should start with the book Tidy R programming with the OMOP common data model before reading this one.

## *How is the book organised?*

| Package         | Version | Link |
|-----------------|---------|------|
| DrugUtilisation | 1.1.0   |      |
| omock           | 0.6.2   |      |

## **How is the book organised?**

TO ADD

## **Citation**

TO ADD

## **License**

## **Code**

The source code for the book can be found at this Github repository

## **Packages version**

This book is rendered using the following version of packages:

```
Finding R package dependencies ... Done!
```

Note we only included the packages called explicitly in the book.

# 1. Proposed structure for the book

- Preface
- Introduction
  - Drug Utilisation studies
  - Drug data in the OMOP CDM
  - Synthetic data used
- Drug Users identification
  - Create drug codelists
    - \* Stratify and subset codelists
    - \* Drug combination codelists
    - \* Import and export codelists
  - Diagnostics
    - \* PhenotypeR
    - \* Dose calculation
  - Identify users
    - \* Drug users cohort
    - \* Identify new users
    - \* Concatenation of prescriptions
- Population level DrugUtilisation
  - Denominator cohort
    - \* General Population
    - \* Target Population
    - \* External Denominator

## *1. Proposed structure for the book*

- Incidence
  - \* Standardised Incidence Rate
  - \* Incidence Rate Ratio
  - \* Standardised Incidence Rate Ratio
- Prevalence
  - \* Point Prevalence
  - \* Period Prevalence
- Patient level DrugUtilisation
  - New users characterisation
    - \* Characterisation
    - \* Large Scale Characteristics (Data Driven Characterisation)
  - Indication
    - \* Indication in the data base
    - \* Phenotype Driven indication
    - \* Data Driven indication
    - \* Linked indication
  - Drug use
    - \* How to calculate daily dose?
    - \* Summarise drug use
  - Treatments
    - \* Alternative treatments
    - \* Comedications
  - Discontinuation
    - \* As survival
    - \* Proportion of patients covered
    - \* Multistate analysis
  - Drug switching
    - \* Drug restart and drug switching
    - \* Treatment pathways
    - \* Multi-state

## *1. Proposed structure for the book*

- Dose escalation
  - \* Dose summary
  - \* Dose pathways
  - \* Dose states
- DrugUtilisation study
  - Final considerations
  - Examples of DrugUtilisation Studies
  - Other studies

```
library(DrugUtilisation)
library(omock)
```



## **Part I.**

# **Introduction**

## **DrugUtilisation Studies**

- What is a DrugUtilisation Study in general

## 2. Drug data in the OMOP CDM

In this chapter the organisation

### 2.1. The Drug exposure table

The OMOP CDM is a domain organised Common Data Model. Clinical data (events) are recorded into different Clinical tables depending on its domain: for example, drugs records (Drug domain) are recorded in the Drug Exposure table or condition records (Condition domain).

The *Drug Exposure* table has 23 columns, each row meaning a different record. Above a brief description of the different fields.

- **drug\_exposure\_id** Unique surrogate key for each drug record.
- **person\_id** Identifier for the patient for whom the drug exposure is recorded (foreign key to the *person* table).
- **drug\_concept\_id** Identifier for the drug code, it must be a standard concept in the Drug domain. When a source drug record cannot be mapped, set this field to 0. The concept with the most detailed content is preferred. Concept classes are considered in the following order of precedence: *Marketed Product* > *Branded Pack / Clinical Pack* > *Branded Drug / Clinical Drug* > *Branded Drug Component / Clinical Drug Component* > *Branded Drug Form / Clinical Drug Form* > *Ingredient*.

## 2. Drug data in the OMOP CDM

- **drug\_exposure\_start\_date** The date the drug exposure began. Valid entries include the date a prescription was written, the date it was dispensed, or the date a drug administration was recorded. It can not be missing.
- **drug\_exposure\_start\_datetime** Optional datetime precision for the start. This field is not used for the analyses described in this book.
- **drug\_exposure\_end\_date** The date the drug exposure end. This date must be determined at the ETL stage. In the analyses described in this book if missing it is assumed to be the same than the start date.
- **drug\_exposure\_end\_datetime** Optional datetime precision for the end. This field is not used for the analyses described in this book.
- **verbatim\_end\_date** The end date exactly as it appears in the source data (e.g. a discontinuation date), without any imputation. This field is not used for the analyses described in this book.
- **drug\_type\_concept\_id** Describes the provenance of the exposure record — for example, whether it is a prescription written, a prescription dispensed, a drug administered by a provider, or a patient-reported exposure. Use this field to filter to specific exposure types in analyses. See accepted Type Concepts and the OHDSI vocabulary wiki for descriptions.
- **stop\_reason** The reason a person stopped a medication as it appears in the source (e.g. “regimen completed”, “changed”, “removed”). Often not populated. **This field will be retired in CDM v6.0.**
- **refills** Intended refills at the time a prescription was written. Only relevant for records sourced from written prescriptions.
- **quantity** The amount of drug dispensed or administered.
- **days\_supply** The number of days of supply as recorded in the source prescription or dispensing record. This reflects the *prescribed* supply and may differ from the actual exposure duration. Leave empty if not verbatim in the source — do not calculate from other fields. Negative values are invalid and should lead to the record being dropped or investigated. Values exceeding 365 days warrant review.

## 2. Drug data in the OMOP CDM

- **sig** The verbatim prescription instructions as written by the provider (Latin abbreviations or plain text as found in the source).
- **route\_concept\_id** Standard concept representing the route of administration (e.g. oral, intravenous, topical). See accepted Route concepts.
- **lot\_number** The manufacturer’s lot or batch number for the drug product, if available. Particularly relevant for vaccines and biologics.
- **provider\_id** The provider associated with the drug record (e.g. the prescriber or the administering clinician). The ETL must determine which provider to record when multiple providers are involved (e.g. ordering vs. administering physician).
- **visit\_occurrence\_id** Links the exposure to the visit during which the drug was prescribed, administered, or dispensed. Drug exposures must be explicitly associated with a visit in the source data to populate this field.
- **visit\_detail\_id** Links to a more granular visit record in VISIT\_DETAIL, if available. For example, if a drug was administered during an ICU stay within a broader inpatient visit, **visit\_occurrence\_id** would reference the full hospital stay and **visit\_detail\_id** would reference the ICU stay.
- **drug\_source\_value** The verbatim code or text from the source system (e.g. an NDC, Gemscrip, or BNF code). Retained for traceability; not intended for primary analysis.
- **drug\_source\_concept\_id** The concept ID representing the source value, if the source uses an OMOP-supported vocabulary (e.g. RxNorm, dm+d). This field is **discouraged for analysis** — prefer **drug\_concept\_id** for network studies.
- **route\_source\_value** The verbatim route value from the source data. This is mapped to **route\_concept\_id** for standardisation; retained here for reference.
- **dose\_unit\_source\_value** The verbatim unit of dose from the source data. **This is a legacy field scheduled for deprecation in a future CDM version.** It is included for backward compatibility but is not recommended for analysis.

## 2. Drug data in the OMOP CDM

As a summary the most important fields that are used in the subsequent analysis are: **person\_id**, **drug\_concept\_id**, **drug\_exposure\_start\_date**, **drug\_exposure\_end\_date** and **quantity** with: **drug\_type\_concept\_id** and **drug\_source\_concept\_id** that might be used at the cohort creation stage.

### 2.2. Standard and Source fields

Every clinical entity in OMOP CDM is captured at two levels: the standard concept (**drug\_concept\_id** in the case of drug records) and the original source representation (**drug\_source\_value** and **drug\_source\_concept\_id** in the case of drug records). Analyses (in general) will always rely on standard concepts so definitions are generalisable across different OMOP CDM instances.

To show some numbers we will explore the 27th August 2025 vocabulary version using the **delphi-100k** dataset.

```
library(omock)
library(duckdb)
library(dplyr)
library(ggplot2)

cdm <- mockCdmFromDataset(datasetName = "delphi-100k_5.4", source = "duckdb")
```

Loading required package: DBI

```
[1] 0
```

Note: method with signature 'DBIConnection#Id' chosen for function 'dbExistsTable', target signature 'duckdb\_connection#Id'.  
"duckdb\_connection#ANY" would also be valid

## 2. Drug data in the OMOP CDM

```
cdm
```

```
-- # OMOP CDM reference (duckdb) of Delphi-2M -----
```

```
* omop tables: care_site, cdm_source, cohort_definition, concept,  
concept_ancestor, concept_class, concept_relationship, concept_synonym,  
condition_era, condition_occurrence, cost, death, device_exposure, domain,  
dose_era, drug_era, drug_exposure, drug_strength, fact_relationship, location,  
measurement, metadata, note, note_nlp, observation, observation_period,  
payer_plan_period, person, procedure_occurrence, provider, relationship,  
source_to_concept_map, specimen, visit_detail, visit_occurrence, vocabulary
```

```
* cohort tables: -
```

```
* achilles tables: -
```

```
* other tables: -
```

Drug concepts are 61.6% of the total 9,018,807 concepts. In particular when referring to only Standard concepts (3,513,741) the percentage is slightly reduced to 58.3%. The Drug domain is the one containing more concepts in the OMOP CDM.

```
x <- cdm$concept |>  
  mutate(standard_concept = case_when(  
    standard_concept == "S" ~ "Standard",  
    standard_concept == "C" ~ "Classification",  
    .default = "Non standard"  
  )) |>
```

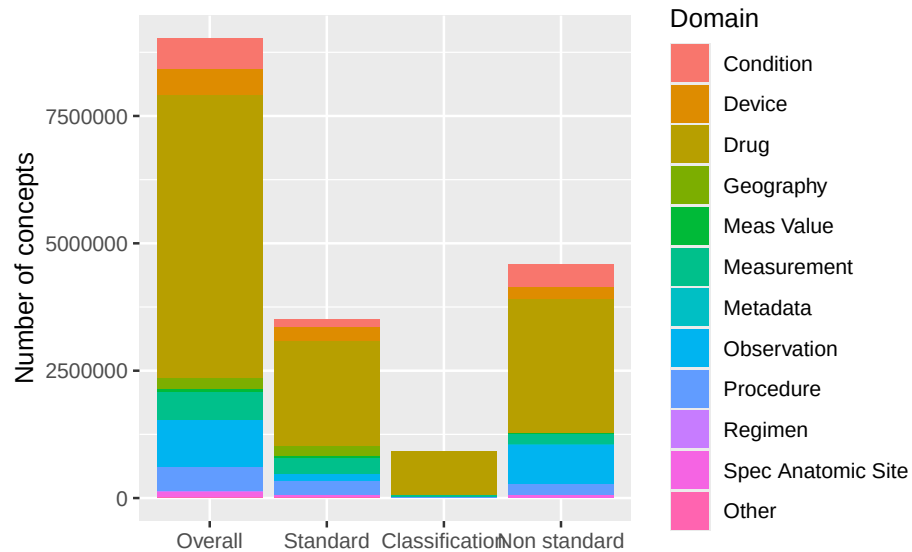
## 2. Drug data in the OMOP CDM

```
group_by(standard_concept, domain_id) |>
tally() |>
collect() |>
mutate(domain_id = if_else(n < 5000, "Other", domain_id)) |>
group_by(standard_concept, domain_id) |>
summarise(n = sum(n), .groups = "drop")
xOverall <- x |>
group_by(domain_id) |>
summarise(n = sum(n), .groups = "drop") |>
mutate(standard_concept = "Overall")
sc <- c("Overall", "Standard", "Classification", "Non standard")
domains <- sort(unique(x$domain_id))
domains <- c(domains[domains != "Other"], "Other")
x <- union_all(x, xOverall) |>
mutate(
  standard_concept = factor(standard_concept, sc),
  domain = factor(domain_id, domains)
)

ggplot(data = x, mapping = aes(x = standard_concept, y = n, fill = domain)) +
  geom_col() +
  labs(x = "", y = "Number of concepts", fill = "Domain")
```



## 2. Drug data in the OMOP CDM



Standard vs. source fields.

### 2.3. Vocabularies

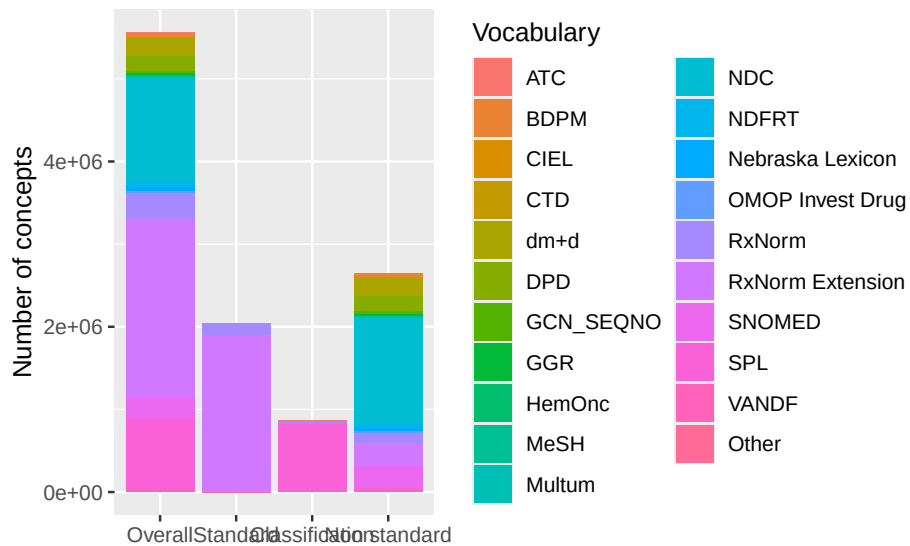
```
x <- cdm$concept |>
  filter(domain_id == "Drug") |>
  mutate(standard_concept = case_when(
    standard_concept == "S" ~ "Standard",
    standard_concept == "C" ~ "Classification",
    .default = "Non standard"
  )) |>
  group_by(standard_concept, vocabulary_id) |>
  tally() |>
  collect() |>
```

## 2. Drug data in the OMOP CDM

```
mutate(vocabulary_id = if_else(n < 5000, "Other", vocabulary_id)) |>
group_by(standard_concept, vocabulary_id) |>
summarise(n = sum(n), .groups = "drop")
xOverall <- x |>
group_by(vocabulary_id) |>
summarise(n = sum(n), .groups = "drop") |>
mutate(standard_concept = "Overall")
sc <- c("Overall", "Standard", "Classification", "Non standard")
vocabs <- sort(unique(x$vocabulary_id))
vocabs <- c(vocabs[vocabs != "Other"], "Other")
x <- union_all(x, xOverall) |>
mutate(
  standard_concept = factor(standard_concept, sc),
  vocabulary = factor(vocabulary_id, vocabs)
)

ggplot(data = x, mapping = aes(x = standard_concept, y = n, fill = vocabulary)) +
  geom_col() +
  labs(x = "", y = "Number of concepts", fill = "Vocabulary")
```

## 2. Drug data in the OMOP CDM



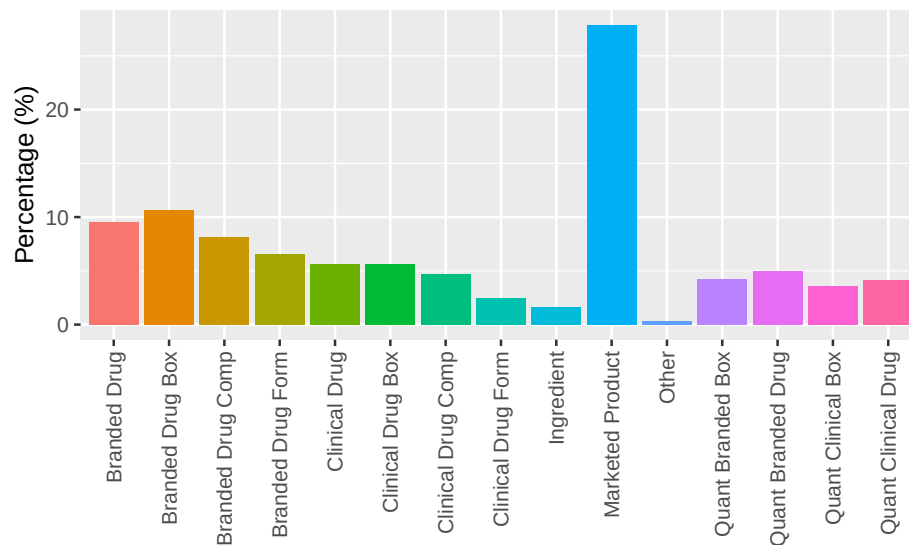
### 2.4. Concept granularly

**Concept granularity.** Drug concepts exist at multiple levels of granularity in the RxNorm/RxNorm Extension hierarchy, from ingredient level up to marketed product. The ETL should map to the most detailed concept available from the source data.

```
x <- cdm$concept |>
  filter(domain_id == "Drug", standard_concept == "S") |>
  group_by(concept_class_id) |>
  tally() |>
  collect() |>
  mutate(concept_class_id = if_else(n < 5000, "Other", concept_class_id)) |>
  group_by(concept_class_id) |>
  summarise(n = sum(n), .groups = "drop") |>
```

## 2. Drug data in the OMOP CDM

```
mutate(n = 100 * n / sum(n))
ggplot(data = x, mapping = aes(x = concept_class_id, y = n, fill = concept_class_id)) +
  geom_col() +
  labs(y = "Percentage (%)", x = "") +
  theme(
    legend.position = "none",
    axis.text.x = element_text(angle = 90, vjust = 0.5, hjust = 1)
  )
```



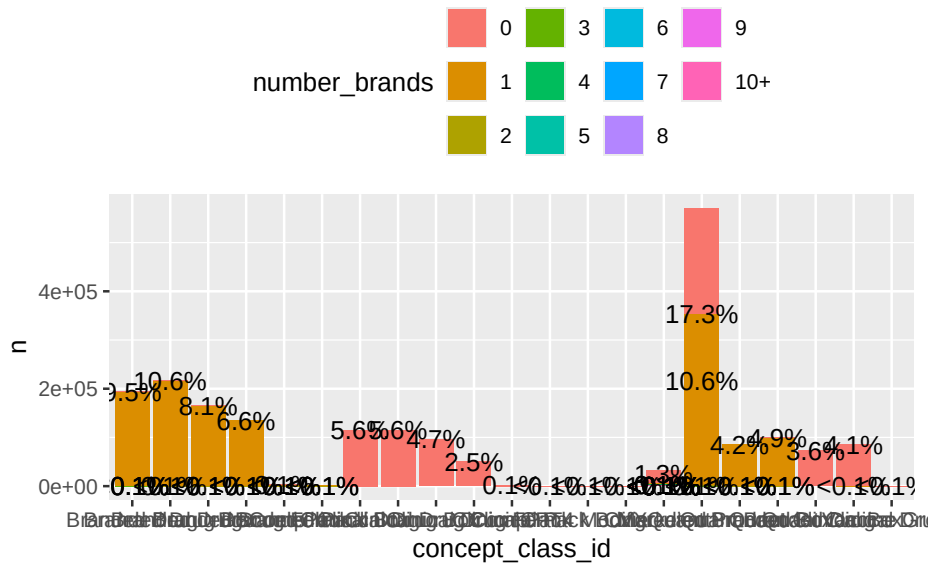
### 2.5. Brand

From the concept id we can deduce the brand of the product

## 2. Drug data in the OMOP CDM

```
x <- cdm$concept |>
  filter(domain_id == "Drug", standard_concept == "S") |>
  inner_join(
    cdm$concept_relationship |>
      filter(relationship_id == "Has brand name") |>
      rename("concept_id" = "concept_id_1"),
    by = "concept_id"
  ) |>
  group_by(concept_id) |>
  summarise(n = n_distinct(concept_id_2)) |>
  right_join(
    cdm$concept |>
      filter(domain_id == "Drug", standard_concept == "S"),
    by = "concept_id"
  ) |>
  mutate(number_brands = coalesce(n, 0)) |>
  mutate(number_brands = if_else(
    number_brands < 10, as.character(as.integer(number_brands)), "10+"
  )) |>
  group_by(number_brands, concept_class_id) |>
  tally() |>
  collect() |>
  mutate(
    number_brands = factor(
      number_brands, c("0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "10+")
    ),
    percentage = sprintf("%.1f", 100 * n / sum(n)),
    percentage = if_else(percentage == "0.0", "<0.1%", paste0(percentage, "%"))
  )
ggplot(data = x, mapping = aes(x = concept_class_id, y = n, fill = number_brands, label = percentage)) +
  geom_col() +
  theme(legend.position = "top") +
  geom_text()
```

## 2. Drug data in the OMOP CDM



For example we can quickly search for the different COVID-19 vaccines and see which are the different brands and number of concepts associated.

```
x <- cdm$concept_ancestor |>
  filter(ancestor_concept_id == 954935) |>
  select("vaccine_concept_id" = "descendant_concept_id") |>
  inner_join(
    cdm$concept_relationship |>
      filter(relationship_id == "Has brand name") |>
      select(
        "vaccine_concept_id" = "concept_id_1",
        "brand_concept_id" = "concept_id_2"
      ),
    by = "vaccine_concept_id"
  ) |>
  inner_join(
```

## 2. Drug data in the OMOP CDM

```
cdm$concept |>
  filter(domain_id == "Drug", standard_concept == "S") |>
  select("vaccine_concept_id" = "concept_id"),
  by = "vaccine_concept_id"
) |>
inner_join(
  cdm$concept |>
    select("brand_concept_id" = "concept_id", "brand" = "concept_name"),
    by = "brand_concept_id"
) |>
group_by(brand) |>
tally() |>
collect()
```

## 2.6. Ingredients

There are 33,218 standard ingredients in this version of vocabularies.

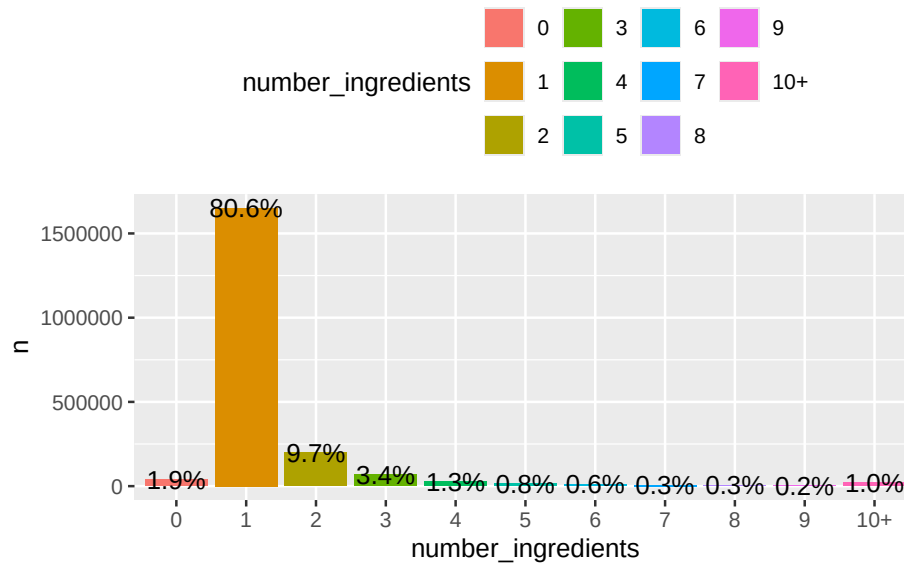
```
x <- cdm$concept |>
  filter(domain_id == "Drug", standard_concept == "S") |>
  inner_join(
    cdm$concept_ancestor |>
      inner_join(
        cdm$concept |>
          filter(concept_class_id == "Ingredient", standard_concept == "S") |>
          select("ancestor_concept_id" = "concept_id"),
          by = "ancestor_concept_id"
        ) |>
      select("concept_id" = "descendant_concept_id", "ancestor_concept_id"),
    by = "concept_id"
  ) |>
```

## 2. Drug data in the OMOP CDM

```
group_by(concept_id) |>
summarise(n = n_distinct(ancestor_concept_id)) |>
right_join(
  cdm$concept |>
    filter(domain_id == "Drug", standard_concept == "S"),
  by = "concept_id"
) |>
mutate(number_ingredients = coalesce(n, 0)) |>
mutate(number_ingredients = if_else(
  number_ingredients < 10, as.character(as.integer(number_ingredients)), "10+"
)) |>
group_by(number_ingredients) |>
tally() |>
collect() |>
mutate(
  number_ingredients = factor(
    number_ingredients, c("0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "10+")
  ),
  percentage = sprintf("%.1f", 100 * n / sum(n)),
  percentage = if_else(percentage == "0.0", "<0.1%", paste0(percentage, "%"))
)
ggplot(data = x, mapping = aes(x = number_ingredients, y = n, fill = number_ingredients)) +
  geom_col() +
  theme(legend.position = "top") +
  geom_text()
```



## 2. Drug data in the OMOP CDM



### 2.7. Daily dose

### 2.8. Combination products

**Combination products.** A single source code representing a combination product may map to multiple standard concept IDs (one per ingredient). In such cases, multiple DRUG\_EXPOSURE rows should be created, one per standard concept.

### 2.9. Routes and drug forms

dose form is only available with RxNorm

## 2. Drug data in the OMOP CDM

There is a minority of concepts (6926, 0.3%) has associated 2 dose forms, most of this records come from an overlap between Injectable Solution, Ijection and Intravenous solution, but this is a minority and in general records will only have one dose (or none) associated.

In Figure XX we can see the distribution of dose forms

```
x <- drugs |>
  left_join(
    cdm$concept_relationship |>
      filter(relationship_id == "RxNorm has dose form") |>
      rename("concept_id" = "concept_id_1"),
    by = "concept_id"
  ) |>
  PatientProfiles::addConceptName(column = "concept_id_2", nameStyle = "") |>
  group_by(concept_id) |>
  summarise(n = n_distinct(concept_id_2)) |>
  right_join(drugs, by = "concept_id") |>
  mutate(number_dose_form = coalesce(n, 0)) |>
  mutate(number_dose_form = if_else(
    number_dose_form < 10, as.character(as.integer(number_dose_form)), "10+"
  )) |>
  group_by(number_dose_form) |>
  tally() |>
  collect() |>
  mutate(
    number_dose_form = factor(
      number_dose_form, c("0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "10+")
    ),
    percentage = sprintf("%.1f", 100 * n / sum(n)),
    percentage = if_else(percentage == "0.0", "<0.1%", paste0(percentage, "%"))
  )

drugs |>
```

## 2. Drug data in the OMOP CDM

```
inner_join(
  cdm$concept_relationship |>
    filter(relationship_id == "RxNorm has dose form") |>
    rename("concept_id" = "concept_id_1"),
  by = "concept_id"
) |>
group_by(concept_id) |>
filter(n_distinct(concept_id_2) == 2) |>
select("concept_id", "concept_id_2") |>
PatientProfiles::addConceptName(column = "concept_id_2", nameStyle = "route") |>
select(!"concept_id_2") |>
collect() |>
arrange(concept_id, route) |>
mutate(id = paste0("route_", row_number())) |>
tidyr::pivot_wider(names_from = "id", values_from = "route") |>
mutate(route = paste0(route_1, "; ", route_2)) |>
group_by(route) |>
tally()

ggplot(data = x, mapping = aes(x = number_dose_form, y = n, fill = number_dose_form,
  geom_col() +
  theme(legend.position = "top") +
  geom_text()
```

number of route categories

### 2.10. Drug eras

**Drug Eras.** The DRUG\_EXPOSURE table stores individual exposure events. The derived DRUG\_ERA table (built post-ETL) collapses these into continuous exposure periods per ingredient, accounting for permissible gaps, and is often more suitable for some summary statistics.

## 2. Drug data in the OMOP CDM

You can build the *Drug Era* using the OmopConstructor R package.

```
library(OmopConstructor)

cdm <- buildDrugEra(cdm = cdm)
```

## Final remarks

*Drug Utilisation Studies in the OMOP Common Data Model* aims to ...

### Learning more

If you find this book useful then joining the **Tidy R in OMOP** OHDSI working group will likely be of interest. Building on many of the concepts and tools seen in this book, the Tidy R in OMOP OHDSI working group aims to (1) create and promote a unified set of resources to guide Tidy R development in OMOP and support the OHDSI community, and (2) establish an overview of available packages relevant to Tidy R programming in OMOP (tidyverse style packages). If you are interested in joining the working group then please email Martí Català or Raivo Kolde.

### Support us

We encourage you to support this work by either citing the book in your papers or documentation, recommending it to your colleagues, or starring the GitHub repository, or simply letting us know how it helped you. Most importantly, please **use it** in research that results in patient benefit.